LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to Aircraft Instruments
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to Aircraft Instruments.
5. Specific Objectives	15.1.1.1 Trainee will be able to explain the various instruments used in an aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of instrument panel layout; Safety: Do not touch live components; Supervision: Instructor monitors trainee interaction with diagrams.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft instrument?
(b) Name one instrument used to measure altitude.
(c) Why are instruments critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of aircraft instruments • Classification of instruments by function • Basic components of flight instruments • Importance of instrument accuracy • Role of instruments in navigation	L & Ds	PPT, IB	15 min
(b)	• Primary flight instruments overview • Secondary and supporting instruments • Instrument grouping and panel layout • Standardization of instrument markings • Basic error sources in instruments	L & Ds	PPT, PDF	20 min
(c)	• Instrument reliability and maintenance basics • Impact of instrument failure on flight • Introduction to instrument redundancy • Future trends in digital instruments • Summary of instrument categories	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between primary and secondary instruments.
(b) Describe the consequences of a primary instrument failure.
(c) List three instruments used for attitude reference.

12. Home Assignment and Follow Up	Prepare a brief note on the elements and mechanisms of aircraft instruments to be covered in the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Instrument elements and mechanism: - Elements, mechanisms, lever mechanisms, gears, hair springs, etc.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min